Performance Analysis of the Vanguard Information Technology ETF (VGT)

Risk-Adjusted Evaluation Against the NASDAQ Composite Index

Asset Management — Group Project
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Abstract

This paper analyses the performance of the Vanguard Information Technology ETF (VGT) relative to the NASDAQ Composite Index over a one-year period from April 2025 to April 2026. Using daily return data obtained from Yahoo Finance, several financial performance measures were calculated, including annualised return, standard deviation, beta, Sharpe ratio, Treynor ratio, Jensen's alpha, correlation, covariance, and tracking error. The findings indicate that VGT achieved higher absolute returns than the benchmark index, but also exhibited higher volatility and market sensitivity. While the ETF generated a positive Jensen's alpha and strong overall growth, the benchmark achieved a slightly higher Sharpe ratio, indicating marginally superior risk-adjusted performance when total risk is considered. Overall, the analysis suggests that VGT may be attractive for growth-oriented investors seeking concentrated exposure to the technology sector, although the higher return potential is accompanied by increased investment risk.

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1. Introduction

Exchange-traded funds (ETFs) have become an increasingly popular investment vehicle in recent years. Their relatively low costs, transparency, and ease of trading make them attractive for both private and institutional investors. In particular, sector-specific ETFs have gained increasing attention, as they allow investors to gain targeted exposure to specific industries or market segments.

In order to evaluate the performance of an investment, market indices are commonly used as benchmarks. These indices reflect the overall development of a specific market or sector and provide a reference point against which individual investments can be assessed. One of the most widely used benchmarks in the technology sector is the NASDAQ Composite Index, which includes a large number of technology-oriented companies.

The Vanguard Information Technology ETF (VGT) is a technology-focused ETF managed by Vanguard and traded on the NYSE Arca exchange. The fund primarily invests in large U.S. technology companies and aims to track the performance of the MSCI US Investable Market Information Technology 25/50 Index (Vanguard, 2026). Its portfolio includes major firms such as Apple, Microsoft, and NVIDIA, which represent a substantial share of the fund's holdings. Due to its strong concentration in the technology sector and its large market capitalisation, VGT is widely used by investors seeking exposure to high-growth technology stocks. Because of differences in portfolio composition, management style, and cost structure, the performance of such ETFs may deviate from that of the broader market index.

When comparing an ETF to its benchmark, it is important to consider not only returns but also the associated level of risk. A higher return may not necessarily indicate better performance if it is accompanied by significantly higher risk. Therefore, financial analysis typically relies on a combination of return and risk measures to evaluate investment performance in a more comprehensive way (Bodie, Kane and Marcus, 2021).

In practice, evaluating investment performance is not always straightforward. Different performance measures may lead to different conclusions, depending on how risk is defined and incorporated. For example, volatility-based measures such as the Sharpe ratio consider total risk, while indicators like beta or the Treynor ratio focus on systematic risk. Using a range of different metrics therefore allows for a more balanced and robust assessment.

The purpose of this paper is to analyse the performance of the Vanguard Information Technology ETF (VGT) over a one-year period using selected financial ratios presented in the lecture. These include average return, standard deviation, beta, the Sharpe ratio (Sharpe, 1966), Jensen's alpha (Jensen, 1968), the Treynor ratio (Treynor, 1965), and the Modigliani–Modigliani measure (Modigliani and Modigliani, 1997). In order to interpret the results, the NASDAQ Composite Index is used as a market benchmark, which allows the ETF's performance to be evaluated within the broader market context, particularly with regard to both return and risk characteristics.

It should be noted that the results of such an analysis depend on the selected time period and data set. Consequently, the findings of this study are limited to the chosen one-year period and may differ under alternative market conditions.

2. Data & Methodology

2.1 Data Source and Time Period

The data used for this analysis was sourced from Yahoo Finance (2026a, 2026b), reflecting daily price histories for the Vanguard Information Technology ETF (VGT) and the NASDAQ Composite Index (IXIC). The sample covers a one-year period from 30 April 2025 to 30 April 2026. Daily closing prices were used to calculate daily returns and the subsequent risk-adjusted performance metrics. A total of 252 trading days were accounted for in the calculations. The risk-free rate of 4.42% applied throughout the analysis is based on the 3-month U.S. Treasury bill rate at the time of data collection (U.S. Department of the Treasury, 2026).

2.2 Performance Metrics

To evaluate the performance of VGT relative to the NASDAQ Composite Index, several financial performance indicators were applied. These measures allow both return and risk characteristics of the ETF to be assessed from different perspectives. Since investment performance cannot be evaluated on the basis of returns alone, the analysis combines traditional return measures with risk-adjusted performance indicators.

Average and annualised return: The arithmetic mean of daily returns was calculated for both VGT and IXIC and annualised over 252 trading days to provide a comparable yearly performance figure. Annualisation allows investment performance to be compared over a standardised one-year horizon. Higher annualised returns generally indicate stronger profitability; however, they must always be interpreted together with risk measures.

Standard deviation: Standard deviation is one of the most widely used measures of total risk in finance. It measures the dispersion of returns around their average value and therefore reflects the volatility of an investment. A higher standard deviation indicates that returns fluctuate more strongly over time, implying greater uncertainty and investment risk. Comparing the standard deviations of VGT and the NASDAQ Composite Index helps determine whether the ETF's higher returns were achieved at the cost of substantially higher volatility.

Beta: Beta measures the systematic risk of an investment relative to the overall market and indicates how sensitive the ETF is to movements in the benchmark index. A beta greater than 1 suggests that the ETF amplifies market movements, while a beta below 1 indicates lower sensitivity. The beta coefficient is particularly relevant within the framework of the Capital Asset Pricing Model (CAPM), where it is used to estimate the expected return of an investment based on its market risk exposure (Bodie, Kane and Marcus, 2021).

Sharpe ratio: The Sharpe ratio evaluates how much excess return an investment generates for each unit of total risk undertaken. It is calculated by subtracting the risk-free rate from the portfolio return and dividing the result by the standard deviation of returns: $SR = (R_p - R_f) / \sigma_p$ (Sharpe, 1966). A higher Sharpe ratio indicates more efficient risk-adjusted performance.

Treynor ratio: The Treynor ratio is another measure of risk-adjusted performance, but unlike the Sharpe ratio it considers only systematic risk measured by beta rather than total volatility: $TR = (R_p - R_f) / \beta_p$ (Treynor, 1965). The Treynor ratio is

particularly appropriate when analysing diversified portfolios, where unsystematic risk is assumed to be minimised through diversification.

Jensen's alpha: Jensen's alpha measures the abnormal return of an investment relative to the return predicted by the CAPM and represents the portion of return that cannot be explained solely by market exposure and systematic risk: $\alpha = R_p - [R_f + \beta \times (R_m - R_f)]$ (Jensen, 1968). A positive alpha indicates outperformance of the CAPM-predicted return; a negative alpha indicates underperformance.

Covariance and correlation: Covariance measures the direction of the relationship between two return series, while correlation standardises this relationship on a scale from −1 to +1. A high positive correlation indicates that the ETF and the benchmark index tend to move in the same direction, which limits the potential diversification benefit of holding both.

Tracking error: Tracking error measures the degree to which an ETF's returns deviate from those of its benchmark index. It is computed as the annualised standard deviation of the difference between VGT and IXIC daily returns. A low tracking error indicates close alignment with the benchmark, while a higher value suggests greater deviation in performance.

Modigliani–Modigliani measure (M^2): The M^2 measure is a risk-adjusted performance metric derived from the Sharpe ratio. Unlike the Sharpe ratio, which is expressed as a ratio, M^2 converts risk-adjusted performance into percentage return terms, adjusting the portfolio to the same level of risk as the market (Modigliani and Modigliani, 1997). A higher M^2 value indicates superior risk-adjusted performance relative to the benchmark.

All calculations were performed in Microsoft Excel using the historical daily return data described above.

3. Results

Table 1 summarises the calculated performance and risk measures for the Vanguard Information Technology ETF (VGT) and the NASDAQ Composite Index (IXIC) over the period from 30 April 2025 to 30 April 2026.

Table 1: Performance and risk metrics, VGT vs. NASDAQ Composite Index (IXIC).

Metric	VGT	NASDAQ Composite (IXIC)
Average daily return	0.1685%	0.1458%
Annualised return	42.46%	36.75%
Standard deviation (annualised)	20.52%	16.80%
Historical beta	1.1675	—
Sharpe ratio	1.8535	1.9247
Treynor ratio	0.3258	—
Jensen's alpha (annualised)	0.29%	—
M^2 ratio	35.55%	—
Correlation with benchmark	0.9555	—
Covariance	0.03293	—
Tracking error (annualised)	6.67%	—
Risk-free rate	4.42%	4.42%

Source: Authors' calculations based on Yahoo Finance daily price data.

4. Interpretation

The results indicate that the Vanguard Information Technology ETF (VGT) outperformed the NASDAQ Composite Index in terms of raw returns during the selected one-year period. VGT achieved an annualised return of approximately 42.46%, compared with 36.75% for the benchmark index. This suggests that the ETF generated higher profitability for investors over the sample period.

However, the higher return was accompanied by a higher level of risk. The standard deviation of VGT was 20.52%, while the NASDAQ Composite Index recorded a lower volatility of 16.80%. This indicates that the ETF experienced larger fluctuations in returns and was therefore more volatile than the market benchmark.

The calculated beta of 1.1675 further confirms that VGT is more sensitive to market movements than the NASDAQ Composite Index. A beta greater than 1 means that the ETF tends to amplify market movements. When the market rises, VGT may increase by a greater percentage, but it may also decline more sharply during market downturns.

The Sharpe ratio of VGT was calculated at 1.8535, while the market benchmark achieved a slightly higher Sharpe ratio of 1.9247. Since the Sharpe ratio measures excess return per unit of total risk, this result indicates that although VGT generated higher returns, the benchmark index provided slightly better risk-adjusted performance when total volatility is considered.

The Treynor ratio of 0.3258 suggests that VGT achieved a positive excess return per unit of systematic risk. This indicates that the ETF was able to compensate investors for the market risk assumed, reflecting an overall satisfactory risk-adjusted performance. However, the measure should be interpreted in a relative framework, meaning that its informational value depends on comparisons with benchmarks or alternative portfolios. Because the Treynor ratio isolates only systematic risk, it relies on the assumption of full diversification.

Jensen's alpha was positive at approximately 0.29% per year, indicating that VGT slightly outperformed the return predicted by the CAPM. A positive alpha generally suggests that the ETF added value beyond what would normally be expected based on market risk alone (Jensen, 1968).

The Modigliani–Modigliani measure (M^2) was calculated at 35.55% for VGT. Compared with the NASDAQ Composite Index's annualised return of 36.75%, this indicates that VGT slightly underperformed the benchmark on a risk-adjusted basis by approximately 1.20 percentage points. Although VGT achieved a higher raw annualised return, the M^2 result shows that once the ETF is adjusted to the same volatility level as the benchmark, its performance becomes slightly weaker. The ETF's additional return was therefore not sufficient to fully compensate investors for the higher level of risk assumed.

The correlation coefficient between VGT and the NASDAQ Composite Index was 0.9555, indicating a very strong positive relationship between the ETF and the market benchmark. This is expected because both investments are heavily concentrated in technology-related companies, which also limits the diversification benefit of combining the two.

Overall, the analysis suggests that VGT provided superior raw returns compared with the NASDAQ Composite Index, but at the cost of higher volatility and market sensitivity. While the ETF demonstrated positive alpha and strong return generation,

the slightly lower Sharpe ratio and M^2 value indicate that its higher returns were partially offset by the additional risk undertaken.

5. Conclusion

This study evaluated the performance of the Vanguard Information Technology ETF (VGT) compared with the NASDAQ Composite Index over a one-year period using several financial performance measures. The results show that VGT outperformed the market benchmark in terms of annualised return, achieving approximately 42.46% compared with 36.75% for the NASDAQ Composite Index. The ETF also generated a positive Jensen's alpha, suggesting that it slightly exceeded the return predicted by its level of market risk.

At the same time, the analysis revealed that VGT carried higher risk than the benchmark. Both the standard deviation and beta values indicated greater volatility and stronger sensitivity to market movements. Although the ETF produced higher returns, the Sharpe ratio showed that the benchmark index achieved marginally better risk-adjusted performance.

Based on the findings, VGT may be considered an attractive investment for investors seeking strong growth and exposure to the technology sector, particularly those willing to accept higher levels of risk. More risk-averse investors may, however, prefer the broader market benchmark due to its relatively more stable risk-adjusted performance. In conclusion, the Vanguard Information Technology ETF demonstrated strong overall performance during the selected period and can be considered a worthwhile investment option for growth-oriented investors, especially during favourable market conditions in the technology sector.

5.1 Limitations of the Study

Several limitations should be considered when interpreting the results of this analysis. First, the study is based on historical data covering only a one-year period. Financial market conditions can change significantly over time, meaning that the results may not be representative of future performance.

Second, the analysis relies on historical volatility and historical beta estimates, which are subject to change depending on market conditions and investor behaviour. As a result, the calculated risk measures may not fully reflect future risk exposure.

Third, the study focuses primarily on quantitative performance indicators and does not incorporate qualitative factors such as macroeconomic developments, monetary policy, geopolitical events, or structural changes within the technology sector itself.

Finally, although the NASDAQ Composite Index provides an appropriate benchmark for technology-oriented investments, VGT tracks the more specialised MSCI US Investable Market Information Technology 25/50 Index, which may naturally lead to deviations in performance and tracking error.

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